PART 1: Short Questions and Answers

1. Python Basics

Q1. What is Python?

Ans: Python is an easy programming language.

Q2. Why is Python used in AI?

Ans: Python is easy to use and has many AI libraries.

Q3. What is a variable?

Ans: A variable is a name used to store data.

Q4. What are the basic data types in Python?

Ans: int, float, string, and boolean.

Q5. What is an integer?

Ans: An integer is a whole number, like 10 or 20.

Q6. What is a float?

Ans: A float is a number with a decimal point, like 3.5.

Q7. What is a string?

Ans: A string is a group of characters written inside quotes.

Q8. What is a list in Python?

Ans: A list stores many values in one variable.

Q9. What is a tuple?

Ans: A tuple stores many values that cannot be changed.

Q10. What is a dictionary?

Ans: A dictionary stores data in key-value pairs.

Q11. What is a loop?

Ans: A loop repeats a block of code.

Q12. What is a function?

Ans: A function is a block of code used to do a task.

Q13. What is indentation in Python?

Ans: Indentation means spaces at the beginning of a line.

Q14. What is a comment in Python?

Ans: A comment is a note in code. It starts with #.

Q15. What is the difference between = and ==?

Ans: = assigns a value. == checks if two values are equal.

Q16. What is a conditional statement?

Ans: It checks a condition and makes a decision.

Q17. What is the use of print()?

Ans: It shows output on the screen.

Q18. What is the use of input()?

Ans: It takes data from the user.

Q19. What is type() in Python?

Ans: It shows the data type of a value.

Q20. What is the difference between break and continue?

Ans: break stops a loop. continue skips the current step.

2. Numpy Basics

Q21. What is NumPy?

Ans: NumPy is a Python library used for working with numbers and arrays.

Q22. What is an array?

Ans: An array stores many values of the same type.

Q23. How do you import NumPy?

Ans:

```
None
import numpy as np
```

Q24. What is the use of np.array()?

Ans: It creates a NumPy array.

Q25. What is a 1D array?

Ans: A 1D array is a simple list of values.

Q26. What is a 2D array?

Ans: A 2D array stores values in rows and columns.

Q27. What is the use of shape?

Ans: It shows the number of rows and columns.

Q28. What is the use of ndim?

Ans: It shows the number of dimensions of an array.

Q29. What is the use of size?

Ans: It shows the total number of elements.

Q30. What is the use of np.mean()?

Ans: It finds the average value.

Q31. What is the use of np.max() and np.min()? Ans: They find the highest and lowest values.

3. Pandas Basics

Q32. What is Pandas?

Ans: Pandas is a Python library used to read, manage, and analyze data.

Q33. Why do we use Pandas?

Ans: We use Pandas to work with tables and datasets.

Q34. How do you import Pandas?

Ans:

```
None
import pandas as pd
```

Q35. What is a DataFrame?

Ans: A DataFrame is a table with rows and columns.

Q36. What is a Series?

Ans: A Series is a one-dimensional data structure.

Q37. What is a CSV file?

Ans: CSV is a file that stores data in rows and columns.

Q38. How do you read a CSV file?

Ans:

```
None
pd.read_csv("data.csv")
```

Q39. What is the use of head()?

Ans: It shows the first five rows by default.

Q40. What is the use of tail()?

Ans: It shows the last five rows by default.

Q41. What is the use of info()?

Ans: It shows information about a dataset.

Q42. What is the use of describe()?

Ans: It shows basic statistical information about data.

Q43. What is the use of shape?

Ans: It shows the number of rows and columns.

Q44. What is the use of columns?

Ans: It shows all column names.

Q45. What is the use of isnull()?

Ans: It checks for missing values.

Q46. What is the use of dropna()?

Ans: It removes rows or columns with missing values.

Q47. What is the use of fillna()?

Ans: It fills missing values.

Q48. What is data filtering?

Ans: Data filtering means selecting data that matches a condition.

Q49. What is the use of loc?

Ans: It selects data using labels.

Q50. What is the use of iloc?

Ans: It selects data using position numbers.

Q51. What is the use of value_counts()?

Ans: It counts how many times each value appears.

Q52. What is the use of unique()?

Ans: It shows all different values in a column.

4. Data Visualization

Q53. What is data visualization?

Ans: Data visualization means showing data using charts and graphs.

Q54. Why do we use data visualization?

Ans: It helps us understand data easily.

Q55. What is Matplotlib?

Ans: Matplotlib is a Python library used to create graphs.

Q56. What is Seaborn?

Ans: Seaborn is a Python library used to create beautiful statistical graphs.

Q57. What is a bar chart?

Ans: A bar chart uses bars to compare different groups.

Q58. What is a histogram?

Ans: A histogram shows how data is spread into groups.

Q59. What is a scatter plot?

Ans: A scatter plot uses dots to show the relationship between two values.

Q60. What is a heatmap?

Ans: A heatmap uses colors to show values in a table.

Q61. What is a distribution?

Ans: Distribution shows how data values are spread.

Q62. What is correlation?

Ans: Correlation shows how two variables are related.

Q63. What is positive correlation?

Ans: When one value increases, the other also increases.

Q64. What is negative correlation?

Ans: When one value increases, the other decreases.

Q65. What is zero correlation?

Ans: It means there is no clear linear relationship between two variables.

Q66. What is a histogram used for?

Ans: It is used to see the distribution of numerical data.

Q67. What is a scatter plot used for?

Ans: It is used to see the relationship between two numerical variables.

Q68. What is the difference between a bar chart and a histogram?

Ans: A bar chart compares categories. A histogram shows numerical data distribution.

Q69. What is the use of plt.show()?

Ans: It displays a graph.

Q70. What is the use of plt.title()?

Ans: It adds a title to a graph.

5. Exploratory Data Analysis (EDA)

Q71. What is EDA?

Ans: EDA means Exploratory Data Analysis. It helps us understand data before training a model.

Q72. Why is EDA important?

Ans: It helps us find patterns, errors, and missing values.

Q73. What do we check in EDA?

Ans: We check missing values, data types, patterns, and relationships.

Q74. What is missing data?

Ans: Missing data means some values are not available.

Q75. What is an outlier?

Ans: An outlier is a value that is very different from other values.

Q76. What is a correlation matrix?

Ans: It is a table that shows the correlation between variables.

Q77. What is the use of a heatmap in EDA?

Ans: It helps us see relationships between variables using colors.

6. Machine Learning Basics and Pipeline

Q78. What is Machine Learning?

Ans: Machine Learning is a way for computers to learn from data.

Q79. What is a Machine Learning model?

Ans: A model is a system that learns patterns from data.

Q80. What is a Machine Learning pipeline?

Ans: A pipeline is a series of steps used to build a Machine Learning model.

Q81. What are the main steps of an ML pipeline?

Ans: Data Collection, Data Cleaning, EDA, Model Training, Model Testing, and Deployment.

Q82. What is Data Collection?

Ans: It means collecting data from different sources.

Q83. What is Data Cleaning?

Ans: It means fixing errors and missing values in data.

Q84. What is Data Exploration?

Ans: It means studying data to find patterns and useful information.

Q85. What is Model Training?

Ans: It means teaching a model using training data.

Q86. What is Model Testing?

Ans: It means checking how well a model works using test data.

Q87. What is Model Deployment?

Ans: It means making a trained model ready for real-world use.

Q88. What is training data?

Ans: Training data is the data used to teach a model.

Q89. What is testing data?

Ans: Testing data is the data used to check a model.

Q90. What is prediction?

Ans: Prediction is the answer given by a trained model.

Q91. What is accuracy?

Ans: Accuracy tells us how many predictions are correct.

Q92. What is the formula for accuracy?

Ans: Accuracy = Correct Predictions / Total Predictions × 100.

Q93. What is a feature?

Ans: A feature is an input column used by a model.

Q94. What is a target variable?

Ans: A target is the output that a model tries to predict.

Q95. What is classification?

Ans: Classification predicts a category, like Yes or No.

Q96. What is regression?

Ans: Regression predicts a number, like house prices.

Q97. What is supervised learning?

Ans: It learns from data that already has correct answers.

Q98. What is unsupervised learning?

Ans: It learns patterns from data without given answers.

Q99. What is overfitting?

Ans: Overfitting happens when a model learns training data too much but performs poorly on new data.

Q100. What is underfitting?

Ans: Underfitting happens when a model cannot learn the patterns properly.

Q101. What is a prediction column?

Ans: It is a column that stores the model's predicted answers.

Q102. What is the difference between actual and predicted values?

Ans: Actual values are the real answers. Predicted values are the model's answers.

Q103. What is a dataset?

Ans: A dataset is a collection of data.

Q104. What is a label?

Ans: A label is the correct answer for a data example.

Q105. What is a train-test split?

Ans: It divides data into training and testing parts.

7. Libraries and Tools

Q106. What is Scikit-learn?

Ans: Scikit-learn is a Python library used for Machine Learning.

Q107. What is Keras?

Ans: Keras is a Python library used to build and train neural networks.

Q108. What is the use of sklearn.model_selection?

Ans: It helps divide data into training and testing parts.

Q109. What is the use of train_test_split()?

Ans: It splits a dataset into training and testing data.

Q110. What is the use of fit()?

Ans: It trains a model using data.

Q111. What is the use of predict()?

Ans: It makes predictions using a trained model.

Q112. What is the use of score()?

Ans: It checks the model's performance.

Q113. What is the use of accuracy_score()?

Ans: It calculates the accuracy of a classification model.

Q114. What is the use of model.compile() in Keras?

Ans: It sets how a model will learn and measure its performance.

Q115. What is the use of model.fit() in Keras?

Ans: It trains the model.

Q116. What is the use of model.evaluate() in Keras?

Ans: It checks the model's performance.

Q117. What is the use of model.predict() in Keras?

Ans: It predicts new results.

PART 2: Multiple Choice Questions (MCQ)

Choose the correct answer. The correct answer is written immediately after each question.

1. Python Basics MCQ

1. Which language is commonly used in AI?

A. HTML

B. Python

C. CSS

D. XML

Answer: B. Python

2. Which symbol is used for comments in Python?

A. //

B. /* */

C. #

D. @

Answer: C. #

3. Which function displays output in Python?

A. input()

B. print()

C. show()

D. display()

Answer: B. print()

4. Which function takes input from the user?

A. print()

B. input()

C. scan()

D. get()

Answer: B. input()

5. Which data type stores whole numbers?

A. float

B. string

C. int

D. bool

Answer: C. int

6. Which data type stores True or False?

A. int

B. bool

C. float

D. string

Answer: B. bool

7. Which symbol is used for multiplication?

A. x

B. *

C. #

D. /

Answer: B. *

8. What is the output of `print(2 + 3)`?

A. 23

B. 6

C. 5

D. Error

Answer: C. 5

9. Which keyword is used to define a function?

A. function

B. fun

C. def

D. define

Answer: C. def

10. Which keyword is used for a condition?

A. if

B. for

C. print

D. import

Answer: A. if

11. Which loop is used to repeat code?

A. if

B. for

C. def

D. return

Answer: B. for

12. Which data structure cannot be changed after creation?

A. List

B. Dictionary

C. Tuple

D. Set

Answer: C. Tuple

13. Which brackets are used for a list?

A. ()

B.

C. []

D.

Answer: C. []

14. What is the output of `type(10)`?

A. float

B. int

C. string

D. bool

Answer: B. int

15. Which keyword is used to import a library?

A. include

B. using

C. import

D. library

Answer: C. import

16. What does `len()` do?

A. Finds the length

B. Finds the average

C. Prints output

D. Deletes data

Answer: A. Finds the length

17. What is the output of `10 == 10`?

A. False

B. True

C. 10

D. Error

Answer: B. True

18. Which keyword stops a loop?

A. continue

B. break

C. skip

D. stop

Answer: B. break

2. NumPy and Pandas MCQ

19. Which library is used for numerical calculations in Python?

A. Pandas

B. NumPy

C. Matplotlib

D. Keras

Answer: B. NumPy

20. What is the common alias of NumPy?

A. pd

B. np

C. plt

D. sns

Answer: B. np

21. Which function creates a NumPy array?

A. np.array()

B. np.list()

C. np.data()

D. np.create()

Answer: A. np.array()

22. Which library is mainly used for data analysis?

A. Pandas

B. Keras

C. HTML

D. CSS

Answer: A. Pandas

23. What is the common alias of Pandas?

A. np

B. pd

C. sns

D. plt

Answer: B. pd

24. Which Pandas structure has rows and columns?

A. Series

B. DataFrame

C. Tuple

D. Array

Answer: B. DataFrame

25. Which Pandas structure is one-dimensional?

A. DataFrame

B. Series

C. Matrix

D. Table

Answer: B. Series

26. Which function reads a CSV file?

A. read_file()

B. read_csv()

C. open_csv()

D. load_data()

Answer: B. read_csv()

27. Which function shows the first five rows?

A. tail()

B. info()

C. head()

D. describe()

Answer: C. head()

28. Which function shows the last five rows?

A. tail()

B. head()

C. shape()

D. mean()

Answer: A. tail()

29. Which property shows rows and columns?

A. size

B. shape

C. info

D. columns

Answer: B. shape

30. Which function shows statistical information?

A. head()

B. describe()

C. tail()

D. print()

Answer: B. describe()

31. Which function checks missing values?

A. isnull()

B. head()

C. mean()

D. sort()

Answer: A. isnull()

32. Which function removes missing values?

A. fillna()

B. dropna()

C. isnull()

D. read_csv()

Answer: B. dropna()

33. Which function fills missing values?

A. dropna()

B. fillna()

C. info()

D. shape

Answer: B. fillna()

34. Which function counts repeated values?

A. unique()

B. value_counts()

C. head()

D. columns

Answer: B. value_counts()

35. Which function selects data by position?

A. loc

B. iloc

C. info

D. shape

Answer: B. iloc

36. Which function selects data by labels?

A. iloc

B. loc

C. tail

D. size

Answer: B. loc

37. What does `df.shape` return?

A. Only column names

B. Number of rows and columns

C. Missing values

D. Average value

Answer: B. Number of rows and columns

3. Data Visualization MCQ

38. Which library is used for plotting graphs?

A. Matplotlib

B. NumPy

C. Pandas

D. Keras

Answer: A. Matplotlib

39. What is the common alias of Matplotlib pyplot?

A. pd

B. np

C. plt

D. sns

Answer: C. plt

40. Which library is used for statistical visualization?

A. Seaborn

B. NumPy

C. Python

D. Keras

Answer: A. Seaborn

41. Which chart is used to compare categories?

A. Histogram

B. Bar chart

C. Scatter plot

D. Heatmap

Answer: B. Bar chart

42. Which chart shows data distribution?

A. Histogram

B. Pie chart

C. Line chart

D. Heatmap

Answer: A. Histogram

43. Which chart uses dots?

A. Bar chart

B. Histogram

C. Scatter plot

D. Heatmap

Answer: C. Scatter plot

44. Which chart uses colors to show values?

A. Bar chart

B. Heatmap

C. Histogram

D. Scatter plot

Answer: B. Heatmap

45. Which chart is useful for showing correlation?

A. Heatmap

B. Bar chart

C. Histogram

D. Pie chart

Answer: A. Heatmap

46. What does a positive correlation mean?

A. Both variables increase together

B. One increases, another decreases

C. No relationship

D. Both are always zero

Answer: A. Both variables increase together

47. What does a negative correlation mean?

A. Both increase together

B. One increases, another decreases

C. No relationship

D. Both are equal

Answer: B. One increases, another decreases

48. Which function displays a graph?

A. plt.show()

B. plt.print()

C. plt.display()

D. plt.graph()

Answer: A. plt.show()

49. Which function adds a title to a graph?

A. plt.name()

B. plt.title()

C. plt.heading()

D. plt.label()

Answer: B. plt.title()

50. Which function creates a histogram in Matplotlib?

A. plt.bar()

B. plt.scatter()

C. plt.hist()

D. plt.heatmap()

Answer: C. plt.hist()

51. Which function creates a scatter plot?

A. plt.scatter()

B. plt.hist()

C. plt.bar()

D. plt.show()

Answer: A. plt.scatter()

52. Which Seaborn function creates a heatmap?

A. sns.barplot()

B. sns.heatmap()

C. sns.histplot()

D. sns.scatterplot()

Answer: B. sns.heatmap()

4. EDA MCQ

53. What does EDA stand for?

A. Easy Data Analysis

B. Exploratory Data Analysis

C. Extra Data Algorithm

D. Electronic Data Analysis

Answer: B. Exploratory Data Analysis

54. When do we perform EDA?

A. After deployment only

B. Before model training

C. After deleting data

D. Never

Answer: B. Before model training

55. What is the main purpose of EDA?

A. To understand data

B. To delete all data

C. To create a website

D. To write code only

Answer: A. To understand data

56. What is an outlier?

A. A normal value

B. A missing value

C. A very different value

D. A column name

Answer: C. A very different value

57. What does a correlation matrix show?

A. File size

B. Relationship between variables

C. Only missing values

D. Row numbers

Answer: B. Relationship between variables

58. Which chart helps identify outliers?

A. Box plot

B. Pie chart

C. Bar chart

D. Line chart

Answer: A. Box plot

59. Which command gives a quick summary of a dataset?

A. df.describe()

B. df.delete()

C. df.close()

D. df.print()

Answer: A. df.describe()

5. Machine Learning MCQ

60. What does ML stand for?

A. Machine Language

B. Machine Learning

C. Main Learning

D. Model Language

Answer: B. Machine Learning

61. What is the first step of an ML pipeline?

A. Model Testing

B. Deployment

C. Data Collection

D. Prediction

Answer: C. Data Collection

62. What comes after Data Collection?

A. Data Cleaning

B. Deployment

C. Prediction

D. Testing

Answer: A. Data Cleaning

63. Which step is used to teach a model?

A. Data Collection

B. Model Training

C. Deployment

D. Data Cleaning

Answer: B. Model Training

64. Which data is used to train a model?

A. Testing data

B. Training data

C. New data only

D. Empty data

Answer: B. Training data

65. Which data is used to check a model?

A. Training data only

B. Testing data

C. Missing data

D. Random text

Answer: B. Testing data

66. What does prediction mean?

A. Removing data

B. Finding an expected answer

C. Cleaning data

D. Collecting data

Answer: B. Finding an expected answer

67. What does accuracy measure?

A. Model speed

B. Correct predictions

C. Data size

D. Number of columns

Answer: B. Correct predictions

68. What is the range of accuracy?

A. 0% to 100%

B. 100% to 200%

C. 1 to 1000

D. 10% to 500%

Answer: A. 0% to 100%

69. What is classification?

A. Predicting categories

B. Predicting only decimal numbers

C. Removing columns

D. Cleaning data

Answer: A. Predicting categories

70. What is regression?

A. Predicting categories

B. Predicting numerical values

C. Deleting rows

D. Creating graphs

Answer: B. Predicting numerical values

71. Which is an example of classification?

A. Predicting house prices

B. Predicting temperature

C. Predicting Yes or No

D. Predicting height

Answer: C. Predicting Yes or No

72. Which is an example of regression?

A. Spam or Not Spam

B. Pass or Fail

C. Predicting house prices

D. Yes or No

Answer: C. Predicting house prices

73. What is a feature in ML?

A. Input variable

B. Final answer

C. Error message

D. Graph title

Answer: A. Input variable

74. What is a target variable?

A. Input data

B. Output to predict

C. Missing value

D. Dataset name

Answer: B. Output to predict

75. What is supervised learning?

A. Learning without answers

B. Learning with labeled data

C. Learning without data

D. Learning only from images

Answer: B. Learning with labeled data

76. What is unsupervised learning?

A. Learning with labeled data

B. Learning without labeled answers

C. Learning from teachers only

D. Learning without a computer

Answer: B. Learning without labeled answers

77. What is overfitting?

A. Model learns training data too much

B. Model learns nothing

C. Model has no data

D. Model is deleted

Answer: A. Model learns training data too much

78. What is underfitting?

A. Model learns too much

B. Model cannot learn patterns properly

C. Model has perfect accuracy

D. Model is deployed

Answer: B. Model cannot learn patterns properly

79. What does deployment mean?

A. Training a model

B. Making a model available for real use

C. Deleting a model

D. Collecting data

Answer: B. Making a model available for real use

80. Which library is used for Machine Learning?

A. Scikit-learn

B. HTML

C. CSS

D. Bootstrap

Answer: A. Scikit-learn

81. Which library is used to build neural networks?

A. Pandas

B. Keras

C. Matplotlib

D. NumPy

Answer: B. Keras

82. Which function trains a Scikit-learn model?

A. predict()

B. fit()

C. show()

D. read()

Answer: B. fit()

83. Which function makes predictions?

A. fit()

B. predict()

C. train()

D. clean()

Answer: B. predict()

84. Which function splits data into training and testing parts?

A. train_test_split()

B. read_csv()

C. describe()

D. accuracy_score()

Answer: A. train_test_split()

85. Which function calculates classification accuracy?

A. mean()

B. accuracy_score()

C. shape()

D. fillna()

Answer: B. accuracy_score()

86. Which library provides train_test_split()?

A. NumPy

B. Pandas

C. Scikit-learn

D. Seaborn

Answer: C. Scikit-learn

87. What is the full form of AI?

A. Artificial Intelligence

B. Automatic Internet

C. Advanced Information

D. Artificial Internet

Answer: A. Artificial Intelligence

6. Diabetes Dataset and Actual vs Predicted MCQ

88. What type of problem is diabetes prediction usually?

A. Classification

B. Regression

C. Sorting

D. Clustering only

Answer: A. Classification

89. In a common diabetes dataset, what does Outcome = 1 mean?

A. No diabetes

B. Diabetes

C. Missing value

D. Normal temperature

Answer: B. Diabetes

90. What does Outcome = 0 mean in a common diabetes dataset?

A. Diabetes

B. No diabetes

C. Error

D. Missing data

Answer: B. No diabetes

91. What is the target column in a common diabetes prediction dataset?

A. Age

B. Glucose

C. Outcome

D. Blood Pressure

Answer: C. Outcome

92. What does actual value mean?

A. Model's guess

B. Real answer

C. Random answer

D. Missing answer

Answer: B. Real answer

93. What does predicted value mean?

A. Real answer

B. Model's guess

C. Dataset name

D. Column name

Answer: B. Model's guess

94. If actual = 1 and predicted = 1, what does it mean?

A. Correct prediction

B. Wrong prediction

C. Missing value

D. Error

Answer: A. Correct prediction

95. If actual = 0 and predicted = 1, what does it mean?

A. Correct prediction

B. Wrong prediction

C. Perfect accuracy

D. Missing value

Answer: B. Wrong prediction

96. If a model makes 90 correct predictions out of 100, what is its accuracy?

A. 50%

B. 70%

C. 90%

D. 100%

Answer: C. 90%

97. Which values are commonly used for binary classification?

A. 1 and 2

B. 0 and 1

C. 5 and 10

D. 10 and 20

Answer: B. 0 and 1

PART 3: Important Practical / Code-Based Questions

These are especially important because your quiz is an AI Lab quiz.

1. Python Code Questions

Q1. Write a Python program to print Hello World.

```
None
print("Hello World")
```

Q2. Write a program to add two numbers.

```
None
a = 10b = 20print(a + b)
```

Q3. Write a program to check even or odd.

```
None
num = 10if num % 2 == 0:    print("Even")else:    print("Odd")
```

Q4. Write a Python program using a for loop.

```
None
for i in range(5):    print(i)
```

Q5. How do you find the length of a list?

```
None
numbers = [10, 20, 30]print(len(numbers))
```

2. NumPy Code Questions

Q6. How do you create a NumPy array?

```
None
import numpy as nparr = np.array([10, 20, 30])print(arr)
```

Q7. How do you find the average of an array?

```
None
print(np.mean(arr))
```

Q8. How do you find the maximum value?

```
None
print(np.max(arr))
```

Q9. How do you find the minimum value?

```
None
print(np.min(arr))
```

Q10. How do you create a 2D array?

```
None
arr = np.array([[1, 2], [3, 4]])print(arr)
```

3. Pandas Code Questions

Q11. How do you import Pandas?

```
None
import pandas as pd
```

Q12. How do you read a CSV file?

```
None
df = pd.read_csv("data.csv")
```

Q13. How do you show the first five rows?

```
None
df.head()
```

Q14. How do you show the last five rows?

```
None
df.tail()
```

Q15. How do you show dataset information?

```
None
df.info()
```

Q16. How do you show statistical information?

```
None
df.describe()
```

Q17. How do you find missing values?

```
None
df.isnull().sum()
```

Q18. How do you remove missing values?

```
None
df.dropna()
```

Q19. How do you fill missing values with zero?

```
None
df.fillna(0)
```

Q20. How do you select one column?

```
None
df["Age"]
```

Q21. How do you filter data where Age is greater than 20?

```
None
df[df["Age"] > 20]
```

Q22. How do you count values in a column?

```
None
df["Outcome"].value_counts()
```

4. Visualization Code Questions

Q23. How do you import Matplotlib?

```
None
import matplotlib.pyplot as plt
```

Q24. How do you create a bar chart?

```
None
plt.bar(["A", "B", "C"], [10, 20, 30])plt.show()
```

Q25. How do you create a histogram?

```
None
plt.hist(df["Age"])plt.show()
```

Q26. How do you create a scatter plot?

```
None
plt.scatter(df["Age"], df["Glucose"])plt.show()
```

Q27. How do you import Seaborn?

```
None
import seaborn as sns
```

Q28. How do you create a heatmap?

```
None
sns.heatmap(df.corr(numeric_only=True), annot=True)plt.show()
```

Q29. How do you add a graph title?

```
None
plt.title("My Graph")
```

Q30. How do you add X and Y axis labels?

```
None
plt.xlabel("Age")plt.ylabel("Glucose")
```

5. Machine Learning Code Questions

Q31. How do you import train_test_split?

```
None
from sklearn.model_selection import train_test_split
```

Q32. How do you split data into training and testing parts?

```
None
X_train, X_test, y_train, y_test = train_test_split(    X, y,
test_size=0.2, random_state=42)
```

Q33. What does test_size=0.2 mean?

Ans: 20% of the data is used for testing and 80% for training.

Q34. What does random_state=42 mean?

Ans: It helps get the same random split again.

Q35. How do you train a model?

```
None
model.fit(X_train, y_train)
```

Q36. How do you make predictions?

```
None
y_pred = model.predict(X_test)
```

Q37. How do you calculate accuracy?

```
None
from sklearn.metrics import accuracy_scoreaccuracy =
accuracy_score(y_test, y_pred)print(accuracy)
```

Q38. What are X and y in Machine Learning?

Ans:

X = Input features.

y = Target or output.

Q39. What does y_test contain?

Ans: Actual answers from the testing data.

Q40. What does y_pred contain?

Ans: Answers predicted by the model.

PART 4: Very Important Differences

These questions are commonly asked in short-question exams.

Topic	Difference
Python vs NumPy	Python is a programming language; NumPy is a library.
NumPy vs Pandas	NumPy works with arrays; Pandas works with tables and data.
Series vs DataFrame	Series is 1D; DataFrame is 2D.
Head vs Tail	Head shows first rows; Tail shows last rows.
Dropna vs Fillna	Dropna removes missing values; Fillna fills them.

Matplotlib vs Seaborn	Both create graphs; Seaborn makes statistical graphs easier.
Bar Chart vs Histogram	Bar chart compares categories; histogram shows numerical distribution.
Histogram vs Scatter Plot	Histogram shows distribution; scatter plot shows relationships.
Training vs Testing	Training teaches the model; testing checks the model.
Actual vs Predicted	Actual is the real answer; predicted is the model's answer.
Classification vs Regression	Classification predicts categories; regression predicts numbers.
Supervised vs Unsupervised	Supervised uses labeled data; unsupervised uses unlabeled data.
Accuracy vs Prediction	Prediction gives an answer; accuracy measures correct answers.
Overfitting vs Underfitting	Overfitting learns too much from training data; underfitting fails to learn enough.

PART 5: Extra Questions Your Teacher May Ask

Q1. What is a library in Python?

Ans: A library is a collection of ready-made code.

Q2. What is a dataset?

Ans: A dataset is a collection of information.

Q3. What is a row?

Ans: A row contains information about one record.

Q4. What is a column?

Ans: A column contains one type of information.

Q5. What is a missing value?

Ans: A missing value means data is not available.

Q6. What is data preprocessing?

Ans: It means preparing data before training a model.

Q7. What is normalization?

Ans: Normalization means changing values into a common range.

Q8. What is a model?

Ans: A model learns patterns from data and makes predictions.

Q9. What is an algorithm?

Ans: An algorithm is a set of steps used to solve a problem.

Q10. What is a neural network?

Ans: A neural network is a model inspired by the human brain.

Q11. What is deep learning?

Ans: Deep learning is a type of Machine Learning that uses neural networks with many layers.

Q12. What is a confusion matrix?

Ans: It is a table that shows correct and wrong predictions.

Q13. What is True Positive?

Ans: The model predicts positive, and the actual answer is also positive.

Q14. What is True Negative?

Ans: The model predicts negative, and the actual answer is also negative.

Q15. What is False Positive?

Ans: The model predicts positive, but the actual answer is negative.

Q16. What is False Negative?

Ans: The model predicts negative, but the actual answer is positive.

Q17. What is an epoch in Keras?

Ans: One epoch means the model has gone through the full training dataset once.

Q18. What is batch size?

Ans: Batch size is the number of data samples processed at one time.

Q19. What is a loss function?

Ans: It measures how wrong a model's predictions are.

Q20. What is an optimizer?

Ans: An optimizer helps a model learn by reducing errors.

PART 6: Last-Minute Revision Sheet

Memorize these before the quiz

Question	Answer
Python comment	#
Python output	print()
Python input	input()
NumPy alias	np
Pandas alias	pd
Matplotlib alias	plt
Seaborn alias	sns
DataFrame	2D
Series	1D
First five rows	head()

Last five rows	tail()
Dataset information	info()
Statistical summary	describe()
Missing value check	isnull()
Remove missing values	dropna()
Fill missing values	fillna()
Count values	value_counts()
Average	mean()
Maximum	max()
Minimum	min()
Bar chart	plt.bar()
Histogram	plt.hist()

Scatter plot	plt.scatter()
Heatmap	sns.heatmap()
Display graph	plt.show()
ML training	fit()
ML prediction	predict()
Split dataset	train_test_split()
Accuracy	accuracy_score()
ML library	Scikit-learn
Neural network library	Keras
Input	X
Output	y
Actual answer	y_test

Predicted answer y_pred

Machine Learning Pipeline: Remember the Order

1. Data Collection
   Collect data.
2. Data Cleaning
   Fix missing and wrong data.
3. Data Exploration (EDA)
   Understand data using charts and statistics.
4. Model Training
   Teach the model.
5. Model Testing
   Check the model.
6. Model Deployment
   Use the model in real life.